Topic 1. Classes and Objects. Static. Final. Enum. BigDecimal. LocalDate. Abstract Classes. Interfaces.

Smart Home Project

A smart home consists of various types of devices. Each device has a model, production date, power consumption, and type, which depends on its placement within the home—whether it is built-in or movable.

Smart Light

A Smart Light is a type of device that supports multiple lighting modes: strong, medium, and dim. It also has a maximum lumen output.

- When the mode is strong, the light operates at its maximum lumens.
- In medium mode, the brightness is reduced by 25%.
- In dim mode, the brightness is reduced by 50%.

Vacuum Cleaner

A Vacuum Cleaner may include mop functionality and has a battery level indicator. All vacuum cleaners have the same maximum cleaning area.

- The battery level decreases by 10% for each square meter of the cleaned area.
- If the vacuum cleaner has mop functionality, the battery consumption rate doubles compared to a vacuum without a mop.

Required Functionalities

Implement at least the following functionalities in the Smart Home project:

1. A method that returns the lumen output based on the selected lighting mode.
2. A method that calculates and returns the cleaned area. The method takes a target area as input, and the vacuum cleaner will stop cleaning if the battery is exhausted before reaching the target.

Requirements

- **Exception Handling**:

- Implement **error handling** where necessary (e.g., negative values, exceeding max hours).

Object-Oriented Principles:

- Use **abstract classes and interfaces** to promote **code reusability** and **flexibility**.